

AVAILABLE FOR CONSULTING & SPECIALIST ENGAGEMENTS

CAIO NAHUEL

SOUSA FAGONDE

</> Senior Developer

✚ Geospatial Specialist

🧠 AI Systems Specialist

📁 Technical Consultant

Aerospace engineer specializing in the intersection of **geospatial intelligence**, **agentic AI systems**, and **cloud-native infrastructure**. Consultant for satellite data pipelines, remote sensing analysis, and autonomous multi-agent architectures.

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Who I Am

Engineering rigor meets domain depth — from orbit mechanics to LLM orchestration.

I am a **senior aerospace engineer and systems developer** with a strong focus on applied geospatial intelligence, autonomous AI architectures, and full-stack spatial data platforms. My work sits at the intersection of satellite remote sensing, machine learning, and modern cloud infrastructure.

As a specialist and consultant, I design and build end-to-end solutions — from satellite data ingestion pipelines using **Sentinel-2 / SAR imagery**, through spatial analysis stacks with **PostGIS + GeoServer**, to multi-agent AI systems orchestrated via **NATS** and running on local LLMs via **Ollama**.

My infrastructure work spans privileged Docker environments with **eBPF tracing**, raw-socket network intelligence, GPU-accelerated LLM inference, and sovereign containerized ecosystems. I architect for production from day one — resilient, observable, and scalable.

I consult on technical strategy, system architecture, remote sensing workflows, and AI integration for clients in

ACADEMIC CREDENTIALS

M.Sc. Aerospace Engineering
Numerical Simulation & Remote Sensing
In Progress

B.Sc. Aerospace Engineering
Orbital Mechanics, Mission Design, SAR Analysis
Completed

B.Sc. Science & Technology
Cross-disciplinary engineering foundations
Completed

CONSULTING FOCUS AREAS

- ✓ Geospatial Systems Architecture
- ✓ Satellite Data Pipelines & Remote Sensing
- ✓ Agentic AI & LLM System Design
- ✓ Cloud-Native Infrastructure (Docker/GPU)
- ✓ Environmental Risk Mapping & Analysis

aerospace, environmental monitoring,
and enterprise intelligence sectors.

02 - CORE DOMAINS

Areas of Expertise

Three deep domains, each reinforcing the other in real-world systems.



Aerospace Engineering

Satellite mission design, orbital mechanics, numerical simulation for atmospheric re-entry, and spacecraft subsystem analysis. Applied to Earth observation mission planning and payload performance modeling.

- Orbital Mechanics
- SAR Analysis
- Mission Design
- Sentinel-2
- SWIR / NIR
- Numerical Sim



Geospatial Intelligence

End-to-end remote sensing pipelines: change detection, landslide susceptibility, fire/flood mapping, NRT monitoring, and point cloud 3D visualization. Expert in spatial ETL and OGC service deployment.

- PostGIS
- GeoServer WMS/WFS
- GDAL
- GeoPandas
- Change Detection
- LiDAR / EPT
- TileServer GL



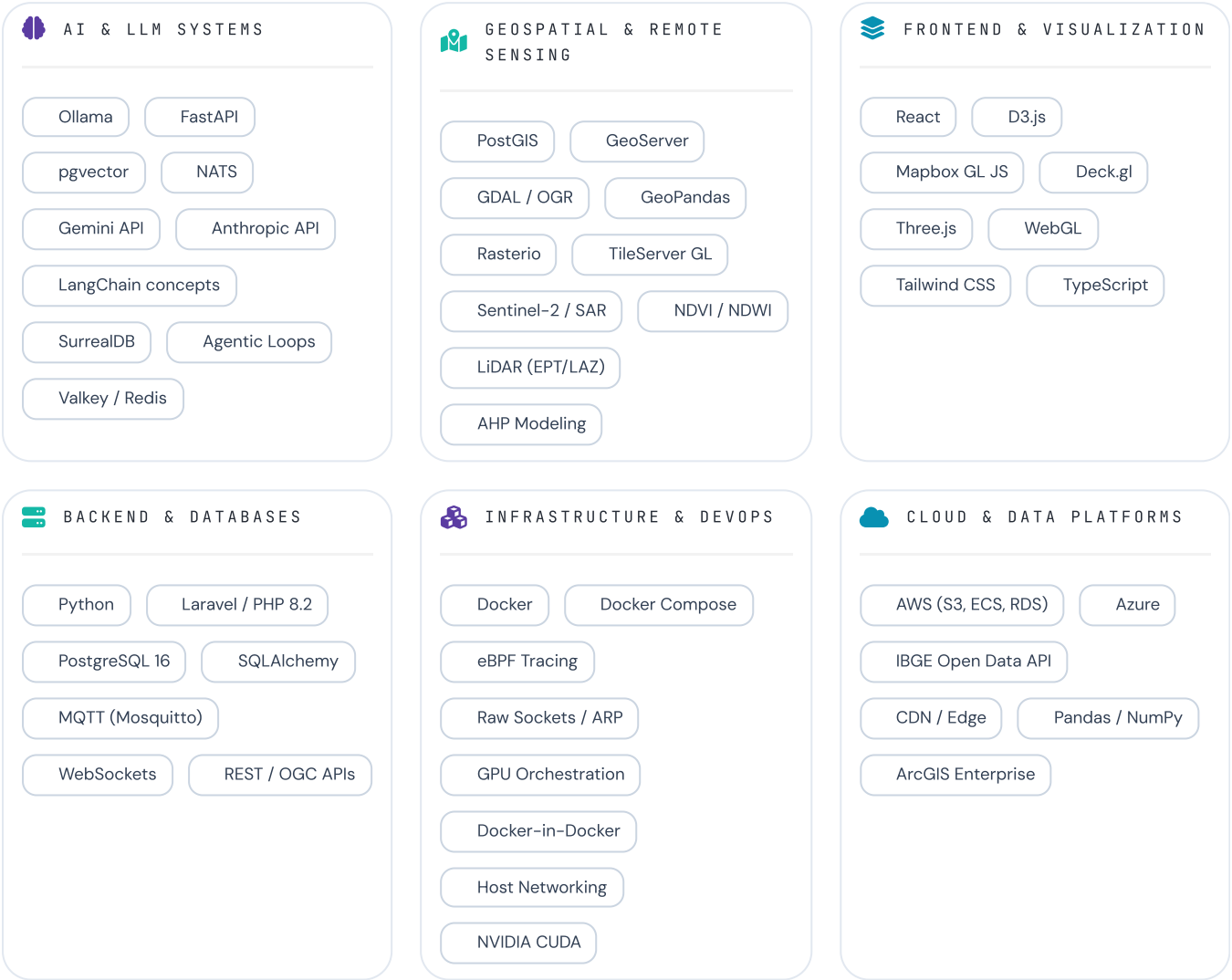
AI Systems & Infrastructure

Agentic AI orchestration with local LLMs, multi-service Docker ecosystems with eBPF instrumentation, RAG pipelines with pgvector, and autonomous NATS-connected multi-agent systems.

- Ollama / LLMs
- FastAPI
- pgvector RAG
- NATS Messaging
- eBPF
- Multi-Agent
- Docker GPU

Technology Stack

The full breadth of tools I architect, deploy, and specialize in.



Key Projects

A selection of systems designed, built, and deployed.

GEOSPATIAL · RISK ANALYSIS

Landslide Susceptibility Mapping

Automated AHP-adaptive pipeline for geological risk categorization in dense urban areas. Fuses slope, lithology, land use, and satellite spectral indices into prioritized risk vectors.

PythonGeoPandasPostGIS
Sentinel-2DEM / Planetscope

GEOSPATIAL · CHANGE DETECTION

Automated Change Detection System

Time-series spectral analysis using NDVI/NDWI index subtraction and feature vector comparison to isolate illegal deforestation and urban expansion with alert generation.

GDALRasterioNDVI/NDWI
PostGIS SAR

GEOSPATIAL · NRT MONITORING

Extreme Events & Fire Monitoring

Near-real-time monitoring of active fire fronts via SWIR band analysis and SAR-based flood extent delimitation independent of cloud cover. Integrated alert pipeline.

SWIR BandsSAR Radar
NRT PipelinePython
GeoServer

AI SYSTEMS · AGENTIC

Project Alfred — Agentic AI OS

Multi-service knowledge OS: FastAPI cortex with host-mode raw socket network intelligence, pgvector RAG memory, Ollama LLM inference, and TileServer offline maps — containerized as a sovereign ecosystem.

FastAPIOllamapgvector
PostGISNATSDocker

GEOSPATIAL · 3D PLATFORMS

IronVision — Point Cloud Platform

Full-stack open geospatial architecture: React + WebGL frontend streaming EPT point clouds directly from S3, GeoServer OGC services, FastAPI ETL orchestration, and PostGIS spatial corrections.

React / WebGLGeoServer
FastAPIPostGISS3 / LAS
D3.js

AI SYSTEMS · INFRASTRUCTURE

Sovereign Forge — AGI Architecture

Autonomous AGI ecosystem: Ollama brain, SurrealDB Zettelkasten memory, eBPF tracepoint instrumentation, Docker-in-Docker Wasm sandbox, MQTT IoT cortex, and Panopticon dashboard — GPU-accelerated.

OllamaSurrealDBeBPF
MQTTDocker GPUValky